

Supplementary material for A prospectively frozen operator for the extent of a source-maintained field on a capacity-limited lattice

Emergent Dynamics Laboratory

This supplement contains the material a reader needs in order to check the main text rather than to take it on trust: what was frozen and when, the term-by-term construction of the tolerance, the individual arms, the arithmetic behind each endpoint, the record the second result is drawn from, and an explicit account of what is missing from this work and why. Every number here is emitted from the same reconciliation file as the main text.

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S1 Provenance, freeze and inertness

S1.1 What was frozen, and when

Two absolute predictions and their ratio were written into a protocol file and hashed before any confirmation arm existed. The frozen predictions carry the status `PRE_RUN` and the methods hash carries `FROZEN`. The methods hash itself is `6676a35fb79d554d99da2b48e69b43f2d3c70f5a0f5cbacb2d258d56a83dd76c`, taken over the files listed in Table S1; the list of files recorded as missing from that hash was empty.

Table S1: The analysis modules hashed into `METHODS_CORE` before any confirmation arm ran, with their SHA-256 truncated to 16 characters. `METHODS_CORE_MISSING` was empty.

File	SHA-256 (first 16)
<code>engine_obtc.py</code>	<code>2172deae5bbabf37</code>
<code>freeze_obfor01.py</code>	<code>f5fd4ad49960a546</code>
<code>guard_obtc.py</code>	<code>ae6f6f241fdf7ae4</code>
<code>kernels_obtr01.py</code>	<code>ad047a9f368f9d9c</code>
<code>kinetics.py</code>	<code>d6b9e24daefd9a9d</code>
<code>lawspec_v2.py</code>	<code>899b7158746d83fa</code>
<code>m6_obfor01.py</code>	<code>fac31046dc188edf</code>
<code>mechanisms_obfor01.py</code>	<code>e441fd4fb2011f36</code>
<code>metrics_obtc.py</code>	<code>8f7d4d5fd920653c</code>
<code>obdi02_protocol.yaml</code>	<code>c65aaef49e07115d</code>
<code>observables_obfor01.py</code>	<code>3eb311202d564eb3</code>
<code>observe.py</code>	<code>55840f35f44ccd42</code>
<code>obtc02_protocol.yaml</code>	<code>65d577460f28d2cc</code>
<code>protocol_obtc02.py</code>	<code>0b39875cef0aea3f</code>
<code>provenance_obfor01.py</code>	<code>a7d1613184802eb7</code>
<code>residual_obfor01.py</code>	<code>270e46a219c47026</code>
<code>run_obfor01.py</code>	<code>b205dfd9aa580484</code>
<code>source_operator.py</code>	<code>537f1d4576c64af7</code>

S1.2 The apparatus does not perturb the system

An instrumented engine and a plain one were run from the same seed for 1500 steps and their full lattice states compared byte for byte. They are identical, with common final state hash

`25b51d2a28943cc6497441fd81d8001be208094f909d1978ae325c903bbdaa86`

so that `STATE_IDENTICAL = TRUE` and `INSTRUMENTATION_ALTERS_THE_LAW = FALSE`. This is a single paired run at a single seed. It is sufficient to falsify interference and insufficient to prove inertness for every trajectory, and we claim only the former.

S1.3 Evidence tiers

Each programme contributing to this work carries one of five tiers, assigned mechanically from what survives in the session that produced this paper rather than from what was once true: `QUALIFIED` (recomputable from surviving hashed bytes and adjudicated in its own programme), `DEVELOPMENTAL` (recorded, but not produced under a freeze that would let it carry a claim), `INVALID` (adjudicated

technically invalid), `LOST_DOCUMENTARY` (the programme ran, its narrative record exists elsewhere, its bytes do not), and `NOT_TESTED`. The assignment is in `provenance/PAPER_SOURCE_BINDING.json`. Every confirmatory statement in the main text rests on `QUALIFIED` evidence only.

S2 The operator and the observable

S2.1 Exact definition of the reported radius

Let w_i be the occupancy of cell i and d_i its toroidal distance to the source cell. Sort the cells by d , form the cumulative sum of w normalised by the total, and take

$$r_q = d \left[\text{searchsorted} \left(\frac{\text{cumsum}(w_{\text{sorted by } d})}{N}, q, \text{side} = \text{left} \right) \right], \quad q = 0.8.$$

Three properties of this estimator matter for Section 5 of the main text and are stated in the source programme’s own definition block: it is a finite-sample quantile of about 120 particles and not a population quantity; it is a *first crossing*, that is a minimum over d , and minima of noisy crossings are biased downwards; and it is supported on the discrete set of achievable toroidal distances.

S2.2 Event order

The within-step order was read from the kinetics module: transport of each species in turn, then reaction, then decay, then feed and outflow. Because reaction runs after transport, a particle born in step t takes no step in step t ; and because every one of the 141009 recorded births across 40 arms occurred at the source cell exactly, with maximum offsets 0 and 0, the point-source idealisation is a checked property rather than an assumption.

S2.3 Where the closure fails

Conditional on the state the operator is exact. Taking expectations fails because the transport factor and the occupancy are functions of the same state. The relevant measured couplings, over 57 arms, are: births equalled the minimum of available substrate and free capacity in 0.866663 of steps, and the source cell had no free capacity in 0.00104094 of them. Hence `CONDITIONAL_EXACT` for the conditional operator, `APPROXIMATE_WITH_CERTIFIED_BOUNDS` for the stationary profile, and `NOT_CLOSED` for the marginal density.

S3 The equivalence margin, term by term

Component	Value (%)
M6_monte_carlo_se_static_percent	0.2754
M6_monte_carlo_se_mobile_percent	0.6024
capacity_certified_error_on_r80_percent	0.0178
intra_step_order_residual_percent	0.2008
historical_arm_to_arm_relative_sd_static_percent	2.1100
historical_arm_to_arm_relative_sd_mobile_percent	4.1520
sampling_se_at_14_arms_static_percent	0.5639
sampling_se_at_14_arms_mobile_percent	1.1097
model error, in quadrature	0.6352
two sampling standard errors	2.2193
declared margin	2.9

The margin is applied to the distance between the observation and the construction’s point prediction. It is not a band around zero. The construction predicted a deficit of -1.235% for the static condition and -5.695% for the mobile one; a tolerance built instead to contain the historical numbers would have had to be about twice as wide as this one.

S4 The confirmation arms

Budget and rules were fixed before any arm ran: 14 arms per condition, a hard cap of 28 in total, and `NO_SEED_REPLACEMENT = TRUE` — an extinction consumes its seed and is never replaced, rerun or deleted. In the event there were 0 extinctions and 0 technically invalid runs, and all 28 arms were analysable.

Table S2: The 28 confirmation arms. Seeds come from a register that excludes every seed appearing anywhere in the historical record. *Final state hash* is the SHA-256 of the full lattice state at the horizon, truncated here to 16 characters; the untruncated values are in `_validation.json`. No arm was rerun, replaced or deleted.

Tag	Final state hash	Seed	r_{80} median	N_X mean	Blocked fraction
S/seed9300000	8f3150a0e420078d	9300000	6.0828	83.04	3.67e-04
S/seed9300001	758204f6b4cc083a	9300001	5.8310	82.80	3.33e-04
S/seed9300002	fab23ac11ce790af	9300002	6.0828	81.73	5.97e-04
S/seed9300003	713206a35f749f8f	9300003	6.0828	81.13	3.00e-04
S/seed9300004	59b40d0d16ea9d7a	9300004	6.0000	81.72	2.61e-04
S/seed9300005	15f2f6317702d4e1	9300005	5.9155	78.35	4.28e-04
S/seed9300006	1c4e79a14128ab14	9300006	6.0828	83.68	4.79e-04
S/seed9300007	a0db588824129ffb	9300007	6.0000	80.33	4.82e-04
S/seed9300008	b08b849942771646	9300008	6.0828	83.77	3.28e-04
S/seed9300009	0c1254b8d888344d	9300009	6.0000	82.99	3.54e-04
S/seed9300010	74555093a06110e9	9300010	5.8310	82.48	4.34e-04
S/seed9300011	beff0675c95dd378	9300011	6.0828	83.21	5.11e-04
S/seed9300012	d6d79ea20a7051e5	9300012	6.0828	82.96	5.43e-04
S/seed9300013	1494b0fe21e66423	9300013	5.8310	75.87	3.62e-04
M/seed9300014	030d7e2ab9eb5bc1	9300014	8.0000	121.53	4.75e-04
M/seed9300015	8052cfd1cce841f6	9300015	8.3657	109.86	3.48e-04
M/seed9300016	e2df28d0d03f676c	9300016	7.8102	119.10	2.95e-04
M/seed9300017	4464f0d9d40152bb	9300017	8.2462	123.02	4.35e-04
M/seed9300018	dc41ff388d53faa5	9300018	8.0623	122.37	3.86e-04
M/seed9300019	345138fe17f7d7a4	9300019	7.6158	117.82	3.95e-04
M/seed9300020	2a38be681ae6f2ec	9300020	8.0623	122.83	2.68e-04
M/seed9300021	f4f37128078c90a5	9300021	8.0623	125.84	4.64e-04
M/seed9300022	40aeb69a459f08cd	9300022	8.0623	124.65	3.51e-04
M/seed9300023	e3eb5d6c13ab2051	9300023	8.5440	118.95	3.83e-04
M/seed9300024	9c83144b09058e2a	9300024	8.0623	123.37	4.22e-04
M/seed9300025	856d64849239a66e	9300025	8.0623	120.70	4.36e-04
M/seed9300026	c1e4f6a3fd05b8f4	9300026	8.0623	118.82	3.73e-04
M/seed9300027	e6580504dab5bbad	9300027	8.0623	127.22	5.09e-04

S5 Endpoint arithmetic

	Static	Mobile	Ratio
Predicted	6.0076	8.0574	1.3412
Observed	5.9991	8.0771	1.3464
Arms	14	14	—
Across-arm s.d.	0.104076	0.218677	—
Across-arm s.e.	0.0278154	0.0584438	—
Deviation (%)	-0.14	0.24	0.39
95 % lower (%)	-1.05	-1.18	-1.30
95 % upper (%)	0.77	1.67	2.07
Margin (%)	2.9	2.9	2.9
Pass	TRUE	TRUE	TRUE

Unity lies outside the ratio interval (`RATIO_ONE_EXCLUDED = TRUE`). The unit of analysis is the arm throughout; frames, steps, particles and birth events are never counted as replicates.

S6 The historical record and the estimator surrogate

S6.1 The radial profile at every radius

r	mobile (116 arms)			static (3 arms)		
	predicted	observed	z	predicted	observed	z
1.0	0.1127	0.1103	-0.588	0.1832	0.2173	+1.622
1.5	0.1676	0.1680	+0.069	0.2629	0.2889	+1.215
2.0	0.2103	0.2102	-0.014	0.3219	0.3358	+0.409
2.5	0.2844	0.2849	+0.062	0.4194	0.4162	-0.087
3.0	0.3397	0.3434	+0.425	0.4869	0.4711	-0.512
4.0	0.4485	0.4514	+0.271	0.6110	0.5804	-8.904
5.0	0.5642	0.5625	-0.143	0.7274	0.7155	-0.494
6.0	0.6444	0.6440	-0.037	0.7994	0.7826	-0.442
7.0	0.7094	0.7073	-0.192	0.8516	0.8297	-0.564
8.0	0.7712	0.7695	-0.158	0.8958	0.8836	-0.317
9.0	0.8219	0.8200	-0.195	0.9280	0.9404	+0.681
10.0	0.8624	0.8608	-0.187	0.9507	0.9486	-0.136
12.0	0.9120	0.9085	-0.525	0.9743	0.9747	+0.039
14.0	0.9487	0.9461	-0.628	0.9883	0.9918	+0.844
17.0	0.9776	0.9761	-0.636	0.9967	0.9964	-0.077

The mobile column rests on 116 arms and its largest standardised deviation is 0.64. The static column rests on 3 arms; its largest standardised deviation is 8.904, at a radius where the absolute difference 0.03407 is no larger than at radii whose standardised deviation is below one. A standard error estimated from three arms can collapse by chance, and here it did. The flag recorded by the source programme is scoped `MOBILE_ONLY`, and we report the static cell as not evaluated. It is printed above so that a reader can see exactly what was set aside, not because it supports anything.

S6.2 The record by lattice size

	$L = 36$	$L = 72$	$L = 96$
Arms	41	37	38
Median-rule residual (%)	-5.172	-4.773	-5.343
Mean-rule residual (%)	-0.5752	-0.1014	-1.416

The median-rule deficit is present at all three sizes and the mean-rule deficit is not, on identical frames.

S6.3 The surrogate

The surrogate draws particles independently from the operator’s own stationary law at the observed per-frame counts and applies the identical summary rule, over 42 surrogate arms of 180 frames. It contains no lattice, no transport, no capacity and no source. Its biases are -1.206% and -1.412% in the static condition and -1.307% and -0.9512% in the mobile one. Note that in the static condition the mean rule is the *worse* of the two under the surrogate: the mean rule is not unbiased, only less biased where it matters.

S6.4 The four summary cells under three accounts

Cell	Observed (%)	Surrogate (%)	Construction (%)
static, median rule	-1.833	-1.206	-1.235
static, mean rule	-1.395	-1.412	-1.485
mobile, median rule	-5.172	-1.307	-5.695
mobile, mean rule	-0.5752	-0.9512	-1.866

The construction’s weakest cell is the mobile mean, where it predicts -1.866% against -0.5752% observed.

S7 Constructions, ablations and the terms that were dropped

Construction	median	residual (%)	mean residual (%)	within-arm s.d.
M6_MOBILE_full	8.0574	-5.6947	-1.8665	1.6808
M4_ablate_birth_flux_to_constant	8.1660	-4.4237	-0.4453	1.7318
M3_ablate_shared_trajectory	8.3746	-1.9827	-1.1658	0.7663
M2_neither_shared_nor_endogenous	8.4879	-0.6563	-0.3414	0.7643
M6_STATIC_full	6.0076	-1.2352	-1.4852	0.6667
M6_STATIC_poisson_births	6.0117	-1.1679	-1.1507	0.6929

Removing the shared source trajectory collapses the within-arm standard deviation from 1.681 to 0.7663, onto the static width, and the median penalty collapses with it. Flattening the birth flux to a Poisson source of the same mean costs -1.271 percentage points. In the two-by-two design the main effects are -3.74 and -1.299 percentage points with an interaction of +0.05543; the three quantities sum to the total by construction, so their sum is not independent evidence for any of them.

S7.1 Terms measured and then dropped

Term	Size on the reported radius	Status
finite torus	unchanged at every size tested	NEGLIGIBLE
finite horizon	carried inside the construction	NEGLIGIBLE
intra-step order	0.2008 %	NEGLIGIBLE
capacity rejection	0.01777 %	NEGLIGIBLE
endogenous source	retained	QUALIFIED
estimator	retained	QUALIFIED

Capacity is negligible here for a reason that can be stated in one line: over a mean lifetime a particle is offered 25.5557 hops and expects 0.00908288 refusals, so 0.990917 of particles are never refused at all, and removing that fraction of moves while concentrating every removal adversarially on outward moves changes the reported radius by 0.01777 %. The birth flux, by contrast, is over-dispersed at a variance-to-mean ratio of 1.285 against 1 for a Poisson source, and it is retained.

S8 What is missing from this work, and why

Evidence	Status	Consequence for this paper
Two earlier programmes on a different question	LOST_DOCUMENTARY	Their raw archives did not survive the container resets during which this paper was written. No quantity, figure, table or abstract sentence draws on them; they appear in Section 7 of the main text as project history with no number attached.
A named earlier manuscript package	NOT_PRESENT	No file in the repository is that package. This paper is therefore a companion rather than an extension, and makes no continuity claim against it.
Part of the raw record behind the historical arms	DOCUMENTARY_ONLY	Those arms are described, never used as confirmatory replicates. Every confirmatory statement rests on the 28 fresh arms.
Biological questions of the wider project	NOT_TESTED	No surviving programme tests them, and none is asserted here in any direction.

For the avoidance of doubt, repeated from the main text:

H3_STATUS	NOT_TESTED
REPRODUCTION_STATUS	NOT_TESTED
HEREDITY_STATUS	NOT_TESTED
AUTONOMOUS_COHESION_STATUS	NOT_ESTABLISHED
X_LAWSPEC_BASELINE	UNCHANGED
ARCHITECTURE_CHANGE_NECESSITY	NOT_ESTABLISHED

S9 Reproducing the numbers in this paper

Each of the 213 rows of `provenance/PAPER_NUMERICAL_RECONCILIATION.json` carries a manuscript label, a value, its units, its estimand, its unit of analysis, its evidence tier, the source file and that file's SHA-256, the JSON path within the file, the recomputed value, the rounding rule, and the sections in which the value may appear. The macros used by the main text and this supplement are generated from that file by `code/emit_numbers.py`, so a number cannot appear in the text unless it appears in the reconciliation first. The four main figures are regenerated by `code/make_figures.py`, and each writes a source-data file alongside its PDF.